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Ultrashort channel chemical vapor deposited bilayer WS_2 field-effect transistors

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ABSTRACT

Two-dimensional transition metal dichalcogenides (TMDs) are potential candidates for next generation channel materials owing to their atomically thin structure and high carrier mobility, which allow for the ultimate scaling of nanoelectronics. However, TMDs-based field-effect transistors are still far from delivering the expected performance, which is mainly attributed to their high contact resistance and low saturation velocity (v_{sat}). In this work, we report high-performance short-channel WS_2 transistors based on bandgap engineering. The bilayer WS_2 channel not only shows a higher average field-effect mobility (μ_{FE}) than the monolayer channel but also exhibits excellent metal-Ohmic contact using a regular physical vapor deposition deposited Ni/Au contact, reducing the R_c value to a record low value of 0.38 $k\Omega \cdot \mu m$ without any intentional doping. The bilayer WS_2 device of the 80 nm channel exhibits a high on-state current of 346 $\mu A/\mu m$ at $V_{ds} = 1$ V, near-zero drain-induced barrier lowering, and a record high I_{on}/I_{off} ratio over 10^9. Furthermore, a record high on-state current of 635 $\mu A/\mu m$ at $V_{ds} = 1$ V and a record high v_{sat} of 3.8×10^6 cm/s have been achieved for a shorter 18 nm channel device, much higher than previous WS_2 transistors. This work reveals the intrinsically robust nature of bilayer WS_2 crystals with promising potential for integration with conventional fabrication processes.

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INTRODUCTION

As the dimensions of nanoelectronics continue to shrink and field-effect transistors enter the nanoscale, there will be more technical challenges to extend Moore's law.[1] Exploring new channel materials is very important for future nanoelectronics applications. Two-dimensional (2D) transition metal dichalcogenides (TMDs) semiconducting materials have attracted intense attention in recent years for electronic applications in extremely scaled complementary metal oxide semiconductor (CMOS) transistors mainly owing to their excellent transport properties and atomically thin body nature.[2–4] As a typical representative of TMDs, tungsten disulfide (WS_2) has attracted increasing research interest due to its large bandgap, small effective mass, and high saturation velocity, all of which are vital for low power and high-performance devices.[5,6] To date, most studies of monolayer WS_2 materials have focused on its scalable growth by chemical vapor deposition (CVD) and electronic properties at low (lateral) electric fields.[7–9] For short-channel devices, the on-state current (I_{on}) becomes less dependent on the low field mobility and is proportional to v_{sat}, following $I_{on} = qn_s v_{sat}$.[10] Here, q is the elementary charge unit, and n_s is the carrier density. Hence, achieving a high v_{sat} is crucial for realizing high-performance devices.[3] Despite the progress in developing WS_2-based transistors, such an important quantity has yet to be fully exploited. The main performance limiting factor of previous WS_2 transistors is the high contact resistance from conventional physical vapor deposition (PVD) metal contact, due to the high metal-induced gap states (MIGS) and surface defects from physical damage.[11] Recently, soft metal or semimetallic contacts, such as In, Bi, and Sb, have been adopted to address contact resistance bottlenecks.[12–14] However, these approaches improve only the contact resistance at the cost of not only degradation in device stability and reliability but also process uniformity and compatibility during fabrication.

Recently, transport simulations have demonstrated that bilayer TMDs showed better device performance than monolayer TMDs because of their higher intrinsic mobility and density of states.[15,16] Moreover, bilayer materials are better immune to fabrication-induced damage or interfacial scattering, resulting in smaller effective Schottky barrier heights and relieving Fermi-level pinning.[3,17] Thus, bilayer WS_2 has more potential as a channel material for next-generation

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electronic devices. However, a thorough study of the electrical properties of extremely scaled bilayer WS_2 transistors with high v_{sat} is lacking.

In this work, we fabricated high-performance field-effect transistors based on CVD WS_2 with regular evaporated Ni/Au contact. The bilayer channel shows two times higher average field-effect mobility (μ_{FE}) than the monolayer channel. Furthermore, the bilayer WS_2 short-channel transistors exhibit a recorded high on-state current and ultralow contact resistance reaching 635 μA/μm at a channel length of 18 nm and at $V_{ds} = 1$ V with a low R_c of 0.38 k$\Omega \cdot \mu$m without any intentional doping. A record saturation velocity of $\sim 3.8 \times 10^6$ cm/s at room temperature is also achieved.

RESULTS AND DISCUSSION

Monolayer and bilayer WS_2 were grown on sapphire substrates using low-pressure chemical vapor deposition in a 1-in. diameter quartz tube furnace. A schematic diagram of the CVD apparatus used to grow WS_2 is illustrated in Fig. 1(a). Inside a single heating zone quartz tube furnace, a corundum boat containing 40 mg WO_2 and 1.5 mg KCl mixed powders was loaded in the center of the high-temperature (890 °C) region. The cleaned single-side polished sapphire c-plane sapphire substrate is placed with its face down on the corundum boat, lying immediately above the tungsten source precursor. Here, molten-salt-assisted CVD is applied for the synthesis of WS_2 under relatively mild conditions. KCl can reduce the melting points of metal oxide precursors.[18,19] Furthermore, the halogen element Cl reacts with WO_2 to form the volatile metal oxychloride WO_xCl_y, which can induce a high mass flux of the metal precursor to the substrate.[19] The reaction pressure was maintained at approximately 1700 Pa by adjusting the valve for 15 min. The introduction of hydrogen could promote WO_2 reduction and enhance the interaction between WS_2 and sapphire, which is beneficial for the epitaxial growth of bilayer WS_2.[20] Moreover, H_2 can protect the straight edges of the equilateral triangles of bilayer WS_2. The optical microscope images of WS_2 are presented in Fig. S1. Note that only monolayer WS_2 can be obtained on the sapphire substrate using only WO_2 powder as the tungsten source precursor. Accordingly, in the presence of the metal-halide salt KCl, a high percentage of large bilayer WS_2 triangular single crystals can be simply achieved.

Figures 1(b) and 1(c) show typical atomic force microscopy (AFM) images of the transferred monolayer and bilayer WS_2 films, respectively. The height profiles extracted from Figs. 1(b) and 1(c)

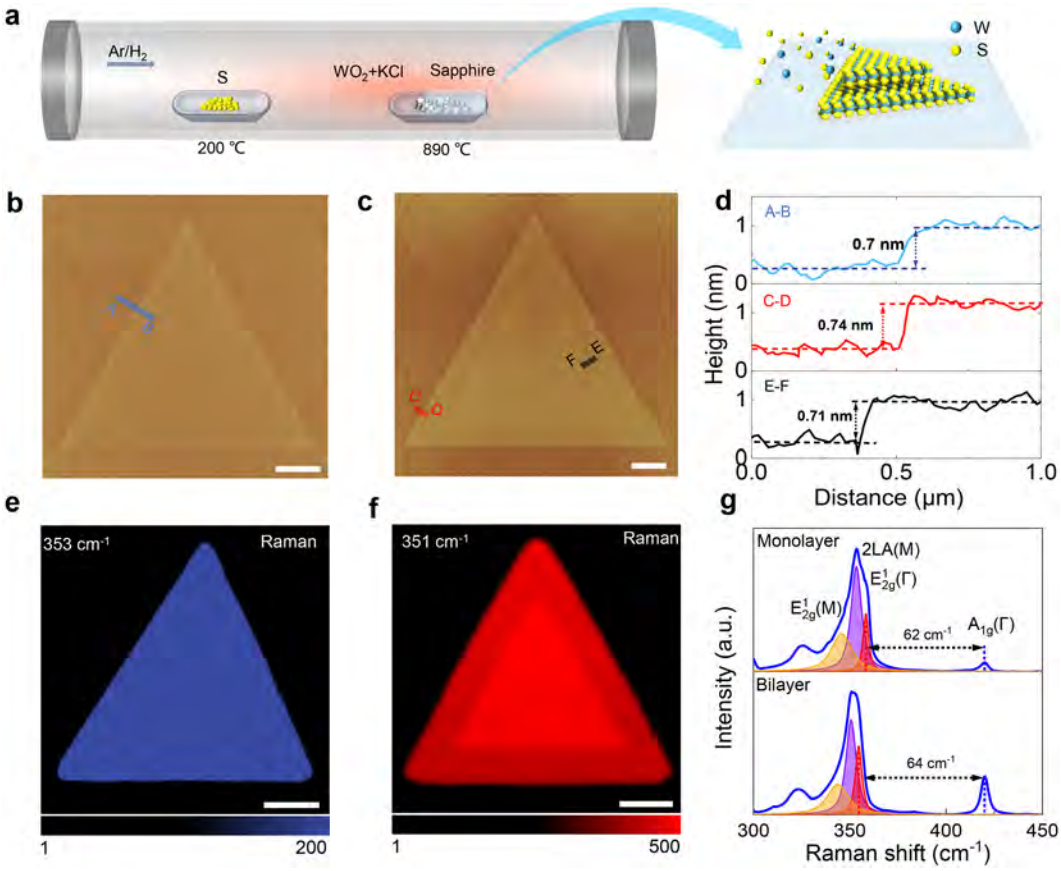


FIG. 1. (a) Schematic of the LPCVD system for the synthesis of WS_2 films. [(b) and (c)] AFM images of transferred monolayer and bilayer WS_2 on 100 nm SiO_2 substrates. Scale bar, (b) 1 μm and (c) 2 μm. (d) Height profile along the lines in [(b) and (c)]. [(e) and (f)] Raman mapping of the transferred monolayer and bilayer WS_2 at 353 and 351 cm^{-1}, respectively. Scale bar, (b) 5 μm and (c) 4 μm. (g) Raman spectra of monolayer and bilayer WS_2.

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confirm that the thickness of single-layer WS_2 is ~0.7 nm, as shown in Fig. 1(d). Cracks and contaminations are barely visible. Moreover, there is no significant color contrast in the whole triangular film, indicating a uniform thickness distribution.[21] Another nondestructive and common accessible method for judging the layer number and crystalline quality is Raman characterization. As shown in Figs. 1(e) and 1(f), Raman intensity mapping can clearly distinguish the spatial shape of the monolayer and bilayer and prove the high uniformity of the transferred WS_2. It is noticeable that the Raman intensity of bilayer WS_2 is stronger than that of monolayer WS_2, which is consistent with the previous reports.[22] Raman spectra excited by the 532 nm laser show rich Raman modes, which are caused by the resonance effect because of the comparable energy of the B exciton of WS_2 and the incident photon, as exhibited in Fig. 1(g).[23] Moreover, we mainly analyze four dominant characteristic Raman peaks, including the first-order optical mode of $E^1_{2g}(M)$, $E^1_{2g}(\Gamma)$, (out-of-plane vibrational mode), and $A_{1g}(\Gamma)$ (in-plane vibrational mode) and the second-order mode of 2LA(M) (longitudinal acoustic mode). The delta values between the $E^1_{2g}(\Gamma)$ and $A_{1g}(\Gamma)$ peaks of monolayer and bilayer WS_2 are 62 and 64 cm^{-1}, respectively, which has also been observed in the previous work.[23] Moreover, the intensity ratio of $2LA(M)/A_{1g}(\Gamma)$ decreases with increasing WS_2 thickness. The ratio is larger than 2 for monolayer WS_2, which provides unique features for identifying monolayer WS_2.[22]

It is known that the bandgap of WS_2 changes with increasing layers, and a direct bandgap semiconductor monolayer WS_2 turns into the direct bandgap semiconductor multilayer WS_2. Photoluminescence (PL) has been shown to serve as a nondestructive and convenient technique to characterize the structures of 2D materials. Figures 2(a) and 2(b) show the PL intensity mapping of the monolayer and bilayer WS_2 at 1.98 eV, respectively. The uniform PL intensity distribution further confirms the high crystalline quality of the transferred CVD WS_2.[21] The PL intensity of monolayer WS_2 is approximately one order of magnitude higher than that of bilayer WS_2, as depicted in Fig. 2(c). Notably, a weak peak located at 1.7 eV can only be observed in bilayer WS_2 due to the indirect bandgap transition, as described in the inset of Fig. 2(c).[5] Kelvin probe force microscopy (KPFM) is an AFM-based technique that is widely used to measure the surface potential and calculate the work function in ambient conditions. More importantly, the high spatial resolution of KPFM is beneficial for the measurement of the work function of 2D materials.[24] The surface potential and work function play critical roles in affecting the performance of 2D transistors. To characterize KPFM, WS_2 films are transferred to a highly oriented pyrolytic graphite (HOPG) substrate. The measured contact potential difference (CPD) is linked to the work function of materials by the following equation (drive routing to sample), $V_{CPD} = (\varphi_{tip} - \varphi_{sample})/q$, where φ_{tip} and φ_{sample} are the work functions of the probe tip and sample, respectively.[24] The mapping surface potential is shown in Fig. 2(d). Brighter parts represent higher values of CPD and work function. The corresponding CPD curve extracted from Fig. 2(d) is shown in Fig. 2(e). The work function of bilayer WS_2 is approximately 40 meV higher than that of monolayer WS_2, which is consistent with the previous reports in the literature.[25] Figure 2(f) represents a schematic image of the energy band structure for the WS_2 layers and Ni electrode based on the surface

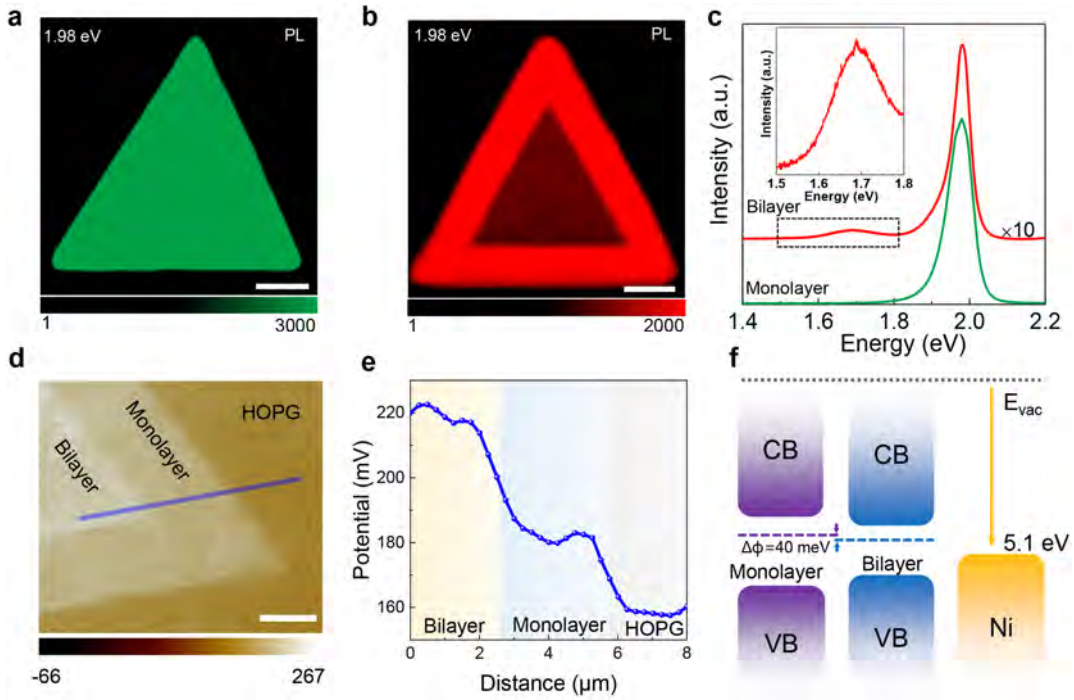


FIG. 2. [(a) and (b)] PL mapping of transferred monolayer and bilayer WS_2 domains at 1.98 eV on 100 nm SiO_2. Scale bar, (a) 5 μm and (b) 4 μm. (c) PL spectra of monolayer and bilayer WS_2 films. (inset) A weak peak located at 1.7 eV for bilayer WS_2. (d) KPFM mapping of bilayer WS_2 on the HOPG substrate. (e) The surface potential difference of the bilayer WS_2 film marked with the color line in (d). (f) Schematic image of the energy band alignment for WS_2 layers and Ni electrodes according to the potential measurement. CB: conduction band. VB: valence band.

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potential result and Fermi level shift. The work functions of monolayer WS_2 and Ni are approximately 4.6 and 5.1 eV, respectively. If the effective mass values for electrons are taken to be the same for monolayer and bilayer WS_2, a conduction band offset (ΔE_C) of ~40 meV at the monolayer–bilayer interface is obtained.[26] Therefore, the ΔE_C of the Ni/bilayer–WS_2 junction is smaller than that of the Ni/monolayer–WS_2 junction, which indicates a smaller effective Schottky barrier height for Ni/bilayer WS_2.[27]

Next, we investigate the changes in the electrical properties for WS_2 transistors by varying the number of layers of channel material and gate dielectrics. Back-gate transistors based on CVD monolayer and bilayer WS_2 are fabricated. Both the channel width and channel length are 1 μm. The channel width of the fabricated WS_2 transistors is measured by scanning electron microscopy (SEM), and the image is shown in Fig. S2. Figures 3(a) and 3(b) display the transfer characteristics and output characteristics of monolayer and bilayer WS_2 on a 100-nm SiO_2 substrate, respectively. We use 1-μm-long devices to minimize the effect of contact resistance on mobility.[28,29] The extracted extrinsic field-effect mobilities [$\mu_{FE} = (g_m L)/(WC_{ox}V_{ds})$, where g_m is the transconductance with upward scans, L and W are the channel length and width, C_{ox} is the gate capacitance, and V_{ds} is the source–drain bias] of monolayer and bilayer WS_2 are 12 and 20.6 $cm^2/V \cdot s$, respectively.[30] It should be noted that the hump of the monolayer WS_2 device with 100 nm SiO_2 could be attributed to the interface traps. The on-state current of bilayer WS_2 FETs is two times higher than that of monolayer WS_2 FETs due to the increased mobility. To improve the performance of the device, high dielectric constant (high-k) La-doped hafnium oxide (HfLaO) is adopted to enhance electrostatic control and dielectric screening.[31] The dielectric constant of HfLaO was estimated by the capacitance-voltage characteristics of metal-oxide-semiconductor (MOS) capacitors in a previous study, which was 26.[32] The gate leakage of the 10 nm HfLaO film at −4 V is $\sim 2 \times 10^{-6}$ A/cm^2, indicating excellent insulator properties (Fig. S3). Meanwhile, compared to thermal SiO_2, atomic layer deposited HfLaO has a low oxide trap density, which is beneficial for decreasing surface charge scattering and improving carrier mobility.[33] As shown in Fig. 3(c), the transfer characteristics of bilayer WS_2 on 10 nm HfLaO show a mobility of 31 $cm^2/V \cdot s$. Moreover, an indistinguishable hysteresis window because of the better dielectric quality is observed (Fig. S4). The larger hysteresis window (~15.5 V) of bilayer WS_2 devices on the 100-nm SiO_2 substrate is shown in Fig. S4.[34] The corresponding on-state current on the 10-nm HfLaO substrate is 58 $\mu A/\mu m$ at $V_{gs} = 4$ V and $V_{ds} = 2$ V, as depicted in Fig. 3(d). Figure 3(e) presents a statistical analysis of the field-effect mobility of monolayer and bilayer WS_2. The positions of the peaks are 15 and 31 $cm^2/V \cdot s$, respectively. To investigate the effective Schottky barrier heights of monolayer and bilayer WS_2 devices, the electrical properties at low temperature are measured. It should be noted that the effective Schottky barrier heights Φ_b of this work are not equivalent to the Schottky barrier heights of conventional metal/semiconductor junctions because the carrier concentration of 2D materials changes with the gate voltage. The corresponding Arrhenius plot of $\ln(I_d/T^{3/2})$ vs $1000/T$ is depicted in Fig. S5. Figure 3(f) shows the effective Schottky barrier height (Φ_b) as a function of gate voltage for monolayer and bilayer WS_2. The effective Schottky

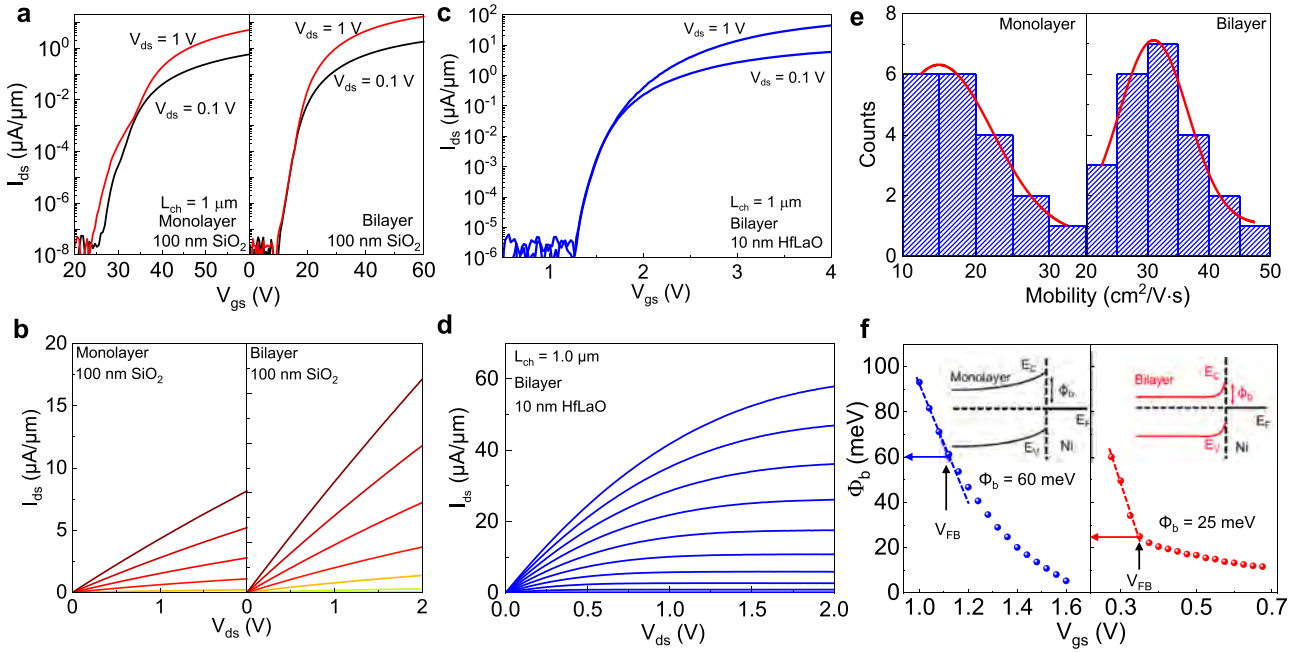


FIG. 3. (a) Comparison of the transfer characteristics of monolayer and bilayer WS_2 FETs on 100-nm SiO_2 substrates with a channel length of 1 μm. (b) Comparison of the output characteristics of monolayer and bilayer WS_2 transistors on 100-nm SiO_2 substrates with a channel length of 1 μm. V_{gt} ($V_{gt} = V_{gs} - V_{th}$) changes from 0 to 40 V with a step of 5 V. [(c) and (d)] Transfer and output characteristics of bilayer WS_2 on 10-nm HfLaO substrates with a channel length of 1 μm. V_{gs} changes from 1 to 4 V with a step of 0.25 V. (e) Statistical distribution of the field-effect mobility of monolayer and bilayer WS_2. The average values are 15 and 31 $cm^2/V \cdot s$, respectively. (f) Extracted effective Schottky barrier heights for the monolayer and bilayer WS_2 nFETs as a function of the gate voltage. Insets show the energy band diagrams for monolayer and bilayer WS_2 transistors.

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barrier height decreases with increasing V_{bg}, which is related to the increase in the electrostatically doped carrier density in WS_2 under the metal contacts.[35] The higher carrier density induced by the positive back gate voltage enhances the carrier concentration in the WS_2 flake, which leads to a smaller effective Schottky barrier.[35,36] To eliminate the effect of tunneling current, Φ_b is extracted at the flatband voltage condition ($V_{gs} = V_{FB}$).[37] The effective Schottky barrier heights of monolayer and bilayer WS_2 are 60 and 25 meV, respectively.

To further evaluate the electrical properties of WS_2 transistors, we fabricate short-channel transistors with a back-gate configuration, as shown in the schematic view in Fig. 4(a). The gate stack includes 10 nm HfLaO gate dielectrics, CVD WS_2 as the channel material, and Ni as the contact metal. Transfer characteristics of transistors based on monolayer and bilayer WS_2 with a channel length of 80 nm at V_{ds} equal to 0.5 and 1 V are shown in Fig. 4(b). Excellent I_{on}/I_{off} ratios of over 10^9 and nearly zero drain-induced barrier lowering (DIBL) are observed for both WS_2 devices. The corresponding output characteristics are shown in Fig. 4(c). The on-state current of monolayer WS_2 is 267 $\mu A/\mu m$ at $V_{ds} = 1$ V, which is mainly limited by the contact resistance.[11] Using bilayer WS_2 as the channel material, the on-state current increases to 346 $\mu A/\mu m$ at $V_{ds} = 1$ V with the same L_{ch} as shown in Fig. 4(d). Figure S6(a) shows the transfer characteristics of bilayer WS_2 transistors at $L_{ch} = 40$ nm. Due to its relatively large bandgap and ultrathin channel, we achieved an excellent I_{on}/I_{off} of $\sim 10^8$. Figure S6(b) presents the corresponding output characteristics with an L_{ch} of 40 nm, exhibiting a high maximum I_{ds} of 490 $\mu A/\mu m$ under $V_{ds} = 1$ V with good saturation behavior. The contact resistance of monolayer and bilayer WS_2 is extracted from the transmission line method (TLM) structure using short channel devices, as shown in Fig. S7.[11] The extracted R_c is significantly reduced from 0.72 $k\Omega \cdot \mu m$ for monolayer WS_2 to 0.38 $k\Omega \cdot \mu m$ for bilayer WS_2. Such an improvement in R_c is attributed to the higher density of states in the bilayer channel. In view of the high mobility and low R_c of bilayer WS_2, we performed further channel length scaling down to 18 nm on bilayer WS_2. A high-resolution cross section transmission electron microscopy (TEM) image of the 18-nm channel length bilayer WS_2 device is shown in Fig. S8. The thickness of the bilayer WS_2 with a clear interface is $\sim$1.5 nm. The transfer characteristics of bilayer WS_2 transistors with 18-nm-long channels are shown in Fig. S9, showing a good on/off ratio ($>10^5$). The corresponding on-state current reaches a record high of 635 $\mu A/\mu m$ at $V_{ds} = 1$ V, as shown in Fig. 4(e). This is mainly attributed to the high mobility bilayer WS_2 channel and ultralow contact resistance. It should be noted that the 18-nm-long device shows less than two times the current increase compared with that of the 80-nm-long device, which is mainly limited by contact resistance.[11] Therefore, much effort is needed to further decrease the contact resistance. We extracted the subthreshold swing (SS) from the transfer characteristics of bilayer WS_2 with different channel lengths, and the result is shown in Fig. S10. As expected, SS increases with decreasing channel length. The 18 nm WS_2 device exhibited an SS value of 295 mV/decade, which could be decreased by further equivalent oxide thickness (EOT) scaling down, top gate and multigate structures.[38,39] To obtain the saturation velocity v_{sat}, we fit the velocity data in Fig. 4(e) to the Caughey–Thomas model[32,40,41]

$$v_d = \frac{\mu F}{\left[1 + \left(\frac{\mu F}{v_{sat}}\right)^{\gamma}\right]^{1/\gamma}}, \quad (1)$$

where v_d is the carrier velocity, F is the lateral field, μ is the mobility derived from the slope of v_d vs F, and γ and v_{sat} are the two free fitting parameters. It is evident that the fitting curves (the solid line) agree well with the experimental data (scatters) at 300 K, as shown in Fig. 4(f). The saturation velocity is extracted to be 3.8×10^6 cm/s at an electron carrier density of 1.7×10^{12} cm^{-2}. This value is much higher than the previous results, which can be further improved by pulse measurement.[9,42,43]

To further assess the performance of bilayer WS_2 FETs, we compare the R_c of CVD WS_2 transistors in the literature with our results, as shown in Fig. 5(a). The Ni contact on CVD bilayer-WS_2 without intentional source/drain doping in this work has the lowest R_c value of $\sim$0.38 $k\Omega \cdot \mu m$ at $n_s \sim 2 \times 10^{13}$ cm^{-2}.[44–48] Figure 5(b) benchmarks I_{ds} vs L_{ch} at $V_{ds} = 1$ V in this work with other transistors based on the CVD channel. We have realized the shortest channel length and a breakthrough in drain current on WS_2 nFETs. Figure 5(c) plots the electron v_{sat} of several bulk semiconducting materials and typical 2D materials vs their band gaps. In view of the atomically thin body nature (thickness < 2 nm) and suitable bandgaps (1–2 eV), our measurements of v_{sat} in bilayer WS_2 provide promising results for logic transistor applications.

CONCLUSIONS

In summary, this work reports the CVD synthesis of large single-crystal monolayer and bilayer WS_2. A record low R_c of 0.38 $k\Omega \cdot \mu m$ is achieved using the regular PVD metal without the doping technique. As a result, the 18 nm L_{ch} WS_2 FETs show a record high I_{ds} of 635 $\mu A/\mu m$ at $V_{ds} = 1$ V and a record high v_{sat} of 3.8×10^6 cm/s at room temperature, which is higher than the previous best results. This work demonstrates the potential of large-area CVD bilayer n-WS_2 for future high-performance short-channel electronic devices.

METHODS

CVD growth

Single-slide polished C/M sapphire (0001) and highly p-doped SiO_2/Si (dielectric oxide 100 nm) substrates were purchased from HeFei Crystal Technical Materials. The sapphire substrates were cleaned sequentially by immersion in acetone, isopropyl alcohol (IPA), and de-ionized (DI) water for 10 min sonication. The SiO_2 substrates were cleaned with a standard RCA-1 procedure. Mixed powders of WO_2 (40 mg, Sigma-Aldrich, 99.98%) and KCl (1.5 mg, Sigma-Aldrich, 99.999%) were placed in the center of the furnace. The S (200 mg, Sigma-Aldrich, 99.99%) powders were placed on the upper stream of the furnace. The furnace was gradually heated to a temperature of 890 °C with 100 sccm Ar. After heating to 890 °C, Ar/H_2 (105/10 sccm) was purged into the tube for 15 min.

Transfer of WS_2 films

The monolayer and bilayer WS_2 films grown on sapphire substrates were transferred onto the target substrates by a polymethylmethacrylate (PMMA)-assisted transfer method. The sapphire substrate with WS_2 films was spin coated with PMMA and baked at 120 °C. Then, a piece of thermal release tape (TRT) was laminated. The TRT/PMMA/WS_2 stack was separated from the sapphire in DI water. TRT was released at 120 °C, and PMMA was removed by acetone.

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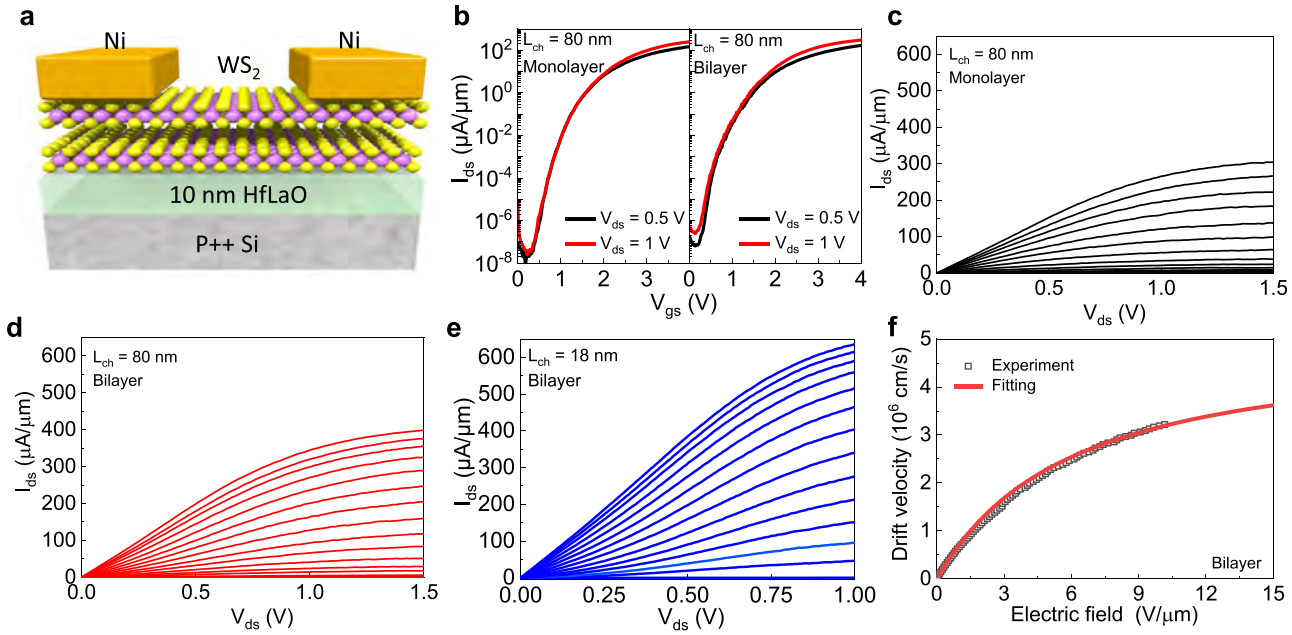


FIG. 4. (a) Schematic diagram of a back-gate bilayer WS_2 transistor on 10-nm HfLaO substrates. (b) Comparison of the transfer characteristics of monolayer and bilayer WS_2 FETs on 10-nm HfLaO substrates with a channel length of 80 nm. The measurement range of the drain and gate current is 100 pA. [(c) and (d)] Corresponding output characteristics of monolayer and bilayer WS_2 transistors on 10-nm HfLaO substrates. V_{gs} changes from 0 to 4 V with a step of 0.25 V. (e) Output characteristics of a bilayer WS_2 transistor with a channel length of 18 nm. V_{gs} changes from −2 to 4 V with a step of 0.5 V. (f) Drift velocity as a function of the electric field, determined from the data of Fig. 4(e). Symbols: experimental data. Solid line: simulated saturation velocity values. The electron carrier density is 1.7×10^{12} cm^{-2}.

Material characterizations

Raman and PL mapping were performed by a LabRAM HR800 Spectra system under 532 nm wavelength excitation. AFM images of surface topography were obtained by a Shimadzu SPM-9700 system. The KPFM measurement was conducted by an atomic force microscope (Dimension Icon, Bruker). The cross-sectional structure of the device was characterized by TEM (Thermo Scientific Tecnai F20).

Device fabrication and electrical characterization

Back-gate FETs were fabricated in this work. The substrate was low resistance (0.01–0.05 Ω · cm) silicon coupled with thermal 100 nm silicon oxide (SiO_2) or 10 nm HfLaO deposited by atomic layer deposition. Monolayer and bilayer WS_2 domains were transferred onto the substrates, patterned with an e-beam lithography step, and etched using O_2/Ar plasma to define the transistor channels with channel width $W = 1$ μm. The channel length of the fabricated WS_2 transistors is measured by SEM and high-resolution TEM. The monolayer devices are fabricated using separated monolayer WS_2 flakes instead of the perimeter of the triangle region in bilayer flakes. For the bilayer devices, we chose the middle region of the bilayer flake as the channel region and etched the monolayer region around the perimeter of the triangle region. The Ni/Au (20/40 nm thick) was used as the contact metal by e-beam evaporation. Electrical measurements were carried

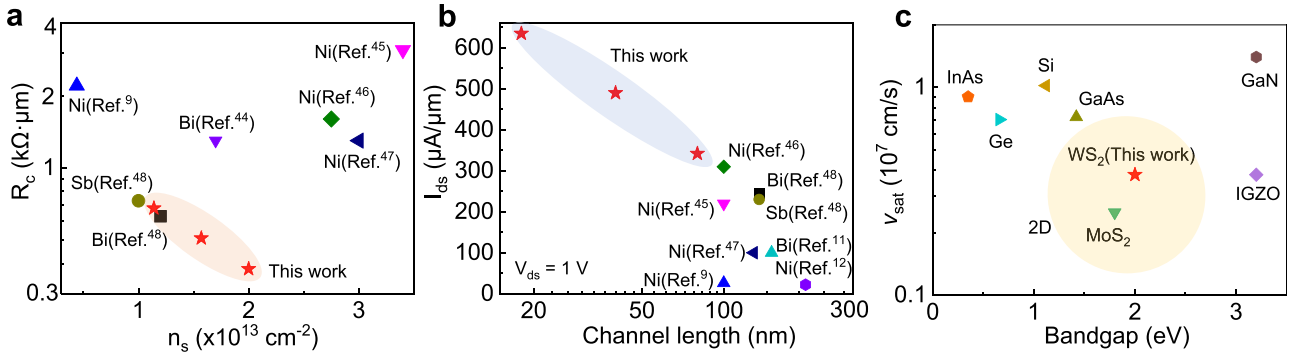


FIG. 5. (a) Benchmarking of the contact resistance in the reported CVD WS_2 FETs. (b). Benchmarking of I_{ds} at $V_{ds} = 1$ V in the reported CVD WS_2 FETs. (c) Electron saturation velocity vs the bandgap for typical bulk semiconductor materials and 2D TMDs at room temperature. We chose two typical n-type TMDs materials (MoS_2 and WS_2). The highest v_{sat} of CVD MoS_2 at room temperature in the literature (Ref. 28) is 2.5×10^6 cm/s. The v_{sat} of bulk materials are adapted from Ref. 28.

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out in a vacuum environment (10^{-5} Torr) in a Lakeshore probe station using a semiconductor parameter analyzer (Agilent B1500A).

SUPPLEMENTARY MATERIAL

See the supplementary material for more experimental details, structure, and electrical characterization.

ACKNOWLEDGMENTS

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Xinhang Shi: Investigation (lead). **Xuefei Li:** Investigation (equal); Supervision (equal); Writing – review & editing (lead). **Qi Guo:** Investigation (supporting). **Min Zeng:** Investigation (supporting). **Xin Wang:** Investigation (supporting). **Yanqing Wu:** Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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